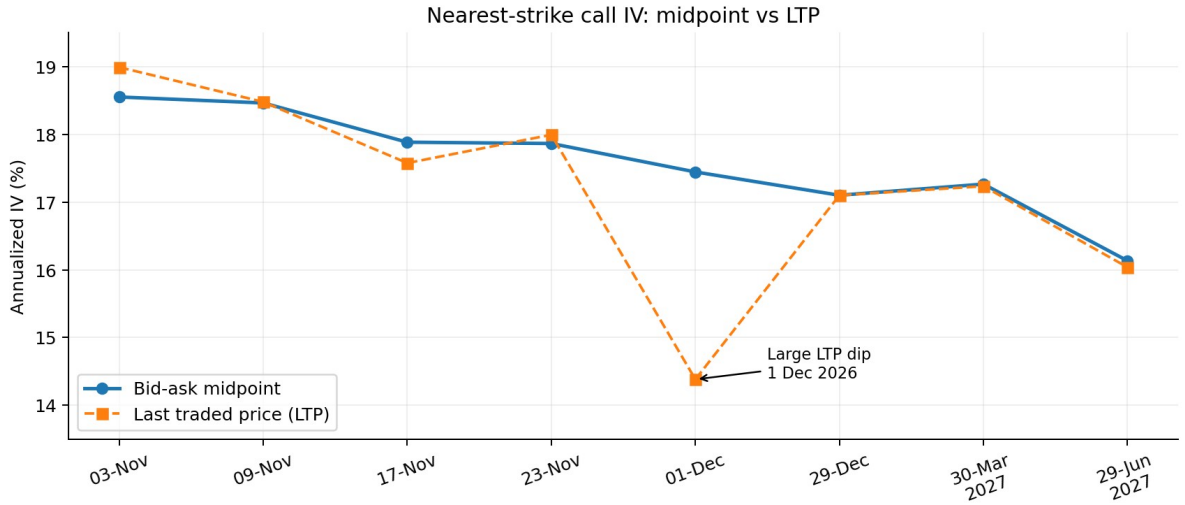
PROJECT REPORT / QUANTITATIVE FINANCE

NIFTY 50 Options: Implied Volatility Estimation and Surface Analysis

Black–Scholes Pricing, Numerical Root-Finding, Volatility Smiles, and Term Structure Analysis



DATASET	MODEL INPUTS	PROJECT SCOPE
8 NSE expiry files	S = 22,753.2 | r = 6.1%	Call-option IV from midpoint and LTP
26 Oct 2026; 3:40 PM IST download	q = 1.52% p.a.	Screened strike and maturity analyses

Report date: 27 October 2026. Prepared from the project source code, original downloaded NSE CSVs, and regenerated numerical diagnostics.

Important: the download timestamp is user-reported; quote-level timestamps are not present in the supplied exported tables. The model's pricing-time setting remains 3:30 PM IST.

Executive summary

This project calculates Black–Scholes implied volatility (IV) for NIFTY 50 European-style index call options using offline NSE option-chain CSV files. It compares IV recovered from the bid–ask midpoint with IV recovered from the last traded price (LTP), then visualizes the strike dependence, expiry dependence, and a triangulated three-dimensional surface.

What was completed

- Parsed eight option-chain downloads and selected calls within ±12% of the spot input.
- Implemented dividend-adjusted Black–Scholes pricing and analytic Vega.
- Inverted market prices with Newton–Raphson; rejected unsolvable quotes.
- Applied transparent screening rules, saved diagnostic CSVs, and produced individual/combined smile plots, ATM-strike term structures, and static/interactive 3D surfaces.

Verified numerical takeaways

Observation	Result
Source rows across 8 expiries	648 strike rows
Rows in project's ±12% strike window	592
Midpoint quotes accepted for plots	355 / 592 (60.0%)
LTP observations accepted for plots	312 / 592 (52.7%)
Median absolute IV difference, 312 accepted pairs	0.10 percentage points
Accepted LTPs outside contemporaneous bid–ask	95 / 312 (30.4%)

Interpretation. The midpoint IV series is usually smoother near the money. Several apparent LTP-IV crashes stem from trades outside the displayed quote interval. The midpoint-based nearest-available-strike term structure broadly falls with maturity, but the longest-dated expiries are represented by only three accepted strikes each.

No claim is made that every option quote was synchronized or that the plotted mesh is an arbitrage-free calibrated surface. The project is an exploratory, reproducible analysis of a single collection session.

1. Research question and scope

Central question

How do NIFTY 50 call-option implied volatilities vary across strike and time to expiry, and how much does the choice of observed option price - bid–ask midpoint versus LTP - change the conclusions?

Specific objectives

- Derive market-implied annualized volatility from observed call prices with a dividend-adjusted Black–Scholes model.
- Compare strike-dependent shapes within each expiry, including downside skew and right-tail curvature.
- Compare nearest-available-strike IV across eight maturity dates.
- Visualize a combined strike–maturity–IV landscape without claiming statistical smoothness or arbitrage-free calibration.
- Understand failures and data limitations using quote-spread, solver, Vega, and LTP-location diagnostics.

Included and excluded

Included now	Not implemented in this version
Call option pricing and IV inversion	Historical/realized volatility estimation
Midpoint and LTP analyses	Put-side empirical IV analysis
Newton–Raphson	Arbitrage-free surface fitting
Screened charts + diagnostic CSVs	Trading signals, forecasting or backtests
Triangulated 3D visualization	Risk-neutral-density extraction

How to read this report

Reported IV is a model-implied parameter that makes the Black–Scholes value match an observed premium; it is not a direct forecast of realized volatility. We label observations as “screened” or “accepted,” not “verified true”: passing simple automated thresholds does not prove a quote is fresh or executable.

2. Data provenance and coverage

Data source: eight user-downloaded NSE NIFTY option-chain CSV exports. Collection time supplied by the project owner: 26 October 2026, 15:40 IST. Each file is named for one expiry. The exported tabular content lists calls to the left of strike and puts to the right; the implemented analysis extracts call-side bid, ask, LTP and trade volume.

The CSVs do not provide individual quote-update timestamps in the extracted fields. The analysis code assumes pricing time of 26 October 2026, 15:30 IST, and an expiry clock time of 15:30. Hence the 3:40 PM collection time and 3:30 PM model timestamp must not be conflated.

Observations by expiry

Expiry	Raw rows	In ±12% window	Mid accepted	LTP accepted
03-Nov-2026	104	104	62	61
09-Nov-2026	104	104	52	48
17-Nov-2026	88	88	54	43
23-Nov-2026	122	109	74	67
01-Dec-2026	77	77	29	24
29-Dec-2026	126	104	78	63
30-Mar-2027	13	3	3	3
29-Jun-2027	14	3	3	3

Totals: 648 raw strike rows; 592 in the strike window; 355 midpoint and 312 LTP observations accepted for their respective plots. The source files contain substantially more strikes for short maturities than for March/June 2027.

Market inputs and conversions

- Underlying spot input S = 22,753.2 index points; not independently matched here to a per-quote timestamp.
- Annual risk-free-rate proxy r = 0.061 (6.1% - MIBOR data: FBIL); annual continuous dividend-yield proxy q = 0.0152 (1.52% - Data: NSE historical data Dividend Yield as per 16th October 2026).
- Each expiry's T is the calendar time from the code's 15:30 snapshot to assumed expiry at 15:30, divided by 365 days.
- Midpoint = (best displayed call bid + best displayed call ask) / 2; LTP is the last recorded trade and can belong to a different moment.

3. Pricing model

Dividend-adjusted Black–Scholes calls and puts

For a European-style call, the project uses the following continuous-dividend-yield form (N is the standard normal cumulative probability function):

CALL: $C = S \exp(-qT) N(d1) - K \exp(-rT) N(d2)$

PUT: $P = K \exp(-rT) N(-d2) - S \exp(-qT) N(-d1)$

$d1 = [\ln(S/K) + (r - q + 0.5 \sigma^2) T] / [\sigma \sqrt{T}]$

$d2 = d1 - \sigma \sqrt{T}$

Symbol	Financial meaning
S, K	Index level and option strike
T	Time to expiry in years
r, q	Continuously compounded rate proxy; dividend yield proxy
sigma	Annualized model volatility (decimal units)
N(·)	Standard normal cumulative distribution function

Why a dividend adjustment?

A dividend-paying index is expected to deliver cash distributions over time. The factor exp(-qT) reduces the present value of the underlying exposure in the call-pricing equation. Omitting q can bias implied volatility, especially at longer maturities. The code uses one fixed annual yield of 1.52% for every expiry; it is a simplification, not a maturity-specific dividend curve.

Vega and put–call parity

The volatility sensitivity used by the root solver is $Vega = S \exp(-qT) \phi(d1) \sqrt{T}$, where phi is the standard normal density. For the same K and T, model-consistent call and put prices satisfy $C - P = S \exp(-qT) - K \exp(-rT)$. Empirical put–call parity is not tested in the current call-only analysis.

Black–Scholes is used as an approximation and numerical benchmark; it assumes a particular diffusion model and constant volatility at the individual-contract pricing step. The observed strike-varying IV is evidence that a single constant model volatility is not sufficient to summarize all market prices.

4. Implied-volatility calculation

The inverse problem

Given a market premium, the unknown is sigma. The algorithm solves the equation f(sigma) = Black–ScholesPrice(sigma) - ObservedMarketPrice = 0. Separate calculations are performed for midpoint and LTP.

Newton–Raphson followed by Brent

Newton update: sigma(next) = sigma(current) - f(sigma(current)) / Vega(sigma(current)). Start from a 20% IV guess. The solver checks the theoretical model price interval before solving, stops only when the price residual is sufficiently small, and switches to bracketed Brent root-finding when Newton would become unstable.

Check	Why it is important
Price within model bounds	Impossible premiums cannot have a valid model IV
Positive volatility bracket	Avoids negative-volatility estimates
Non-negligible Vega	Avoids massive Newton steps for near-insensitive contracts
Small pricing residual	Confirms model premium matches the observation
Failed solve -> NaN	Prevents invented zero-volatility points

A real market-price example

1 December 2026, NIFTY call strike 22,750: bid = INR 536.30, ask = INR 563.55, midpoint = INR 549.925, while LTP = INR 463.35. With the project inputs and time convention, the midpoint-implied IV is 17.4449% and LTP-implied IV is 14.3820%. The numerical routines can invert both quoted premiums, but the inputs reflect different points in the reported quote context.

A numerically converged IV is not the same as a trustworthy market estimate. The premium itself must be considered in context.

5. Quote screening and diagnostics

The project saves every in-range observation to calculated_iv_diagnostics.csv, together with its computed IV (if possible), Vega, spread and plot_status. Charts use rows whose plot_status equals accepted; excluded data remain inspectable.

Rules applied to plots

- Strike within ±12% of S; strike proximity alone does not establish liquidity.
- Market premium must be positive and invertible within the model price bounds.
- Bid/ask must be positive and not crossed; relative spread (ask - bid)/midpoint must be at most 20%.
- Calculated Vega must be at least 20 price points per unit of decimal volatility.
- For plotting only, implied volatility must not exceed 100% annualized; the root solver has a separate 500% numerical bound.

Observed screening outcomes

Screen status	Midpoint	LTP
Accepted	355	312
Missing/nonpositive market price	34	198
Unsolved/outside price bounds	36	14
Wide bid–ask spread	167	64
Invalid bid or ask	0	4

Status categories are hierarchical: an observation is assigned the first relevant rejection reason, so reason counts are not directly comparable as independent tests. Midpoint and LTP statuses can differ even for the same underlying row.

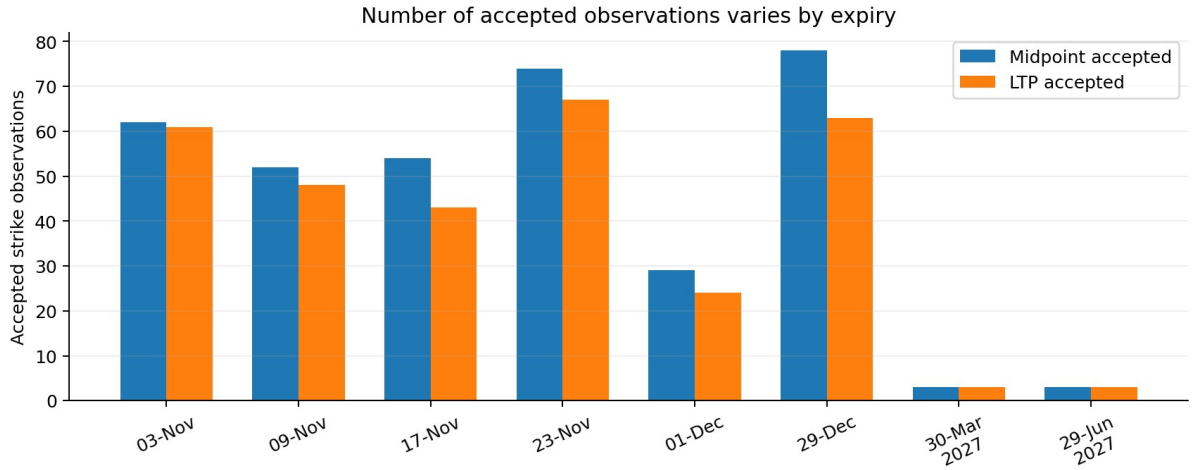


Figure 1. Accepted observations per expiry. Longer-dated curves pass the automated screen at three strikes each, but this remains insufficient coverage for a fully resolved maturity slice.

6. Results: strike-by-strike smiles and skew

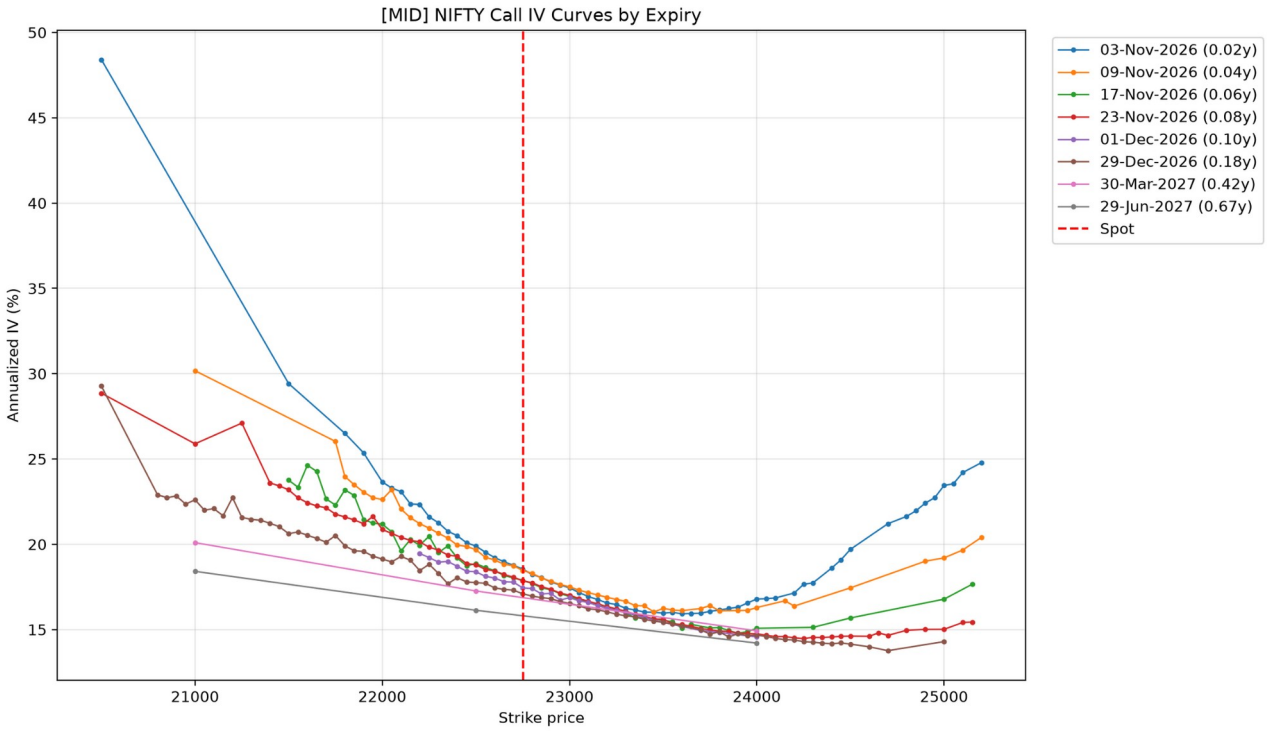


Figure 2. Midpoint-derived NIFTY call IV curves for eight expiries. Vertical dashed line marks S = 22,753.2.

Across short and intermediate expiries, call IV tends to be higher for strikes below spot and lower as strike rises, with some upward turn in far out-of-the-money calls. This is qualitatively compatible with equity-index downside skew and, in selected short maturities, a smile-like right tail.

The magnitude of far-downside IV for deep-in-the-money calls should be treated cautiously: quote freshness, wide absolute spreads, and sensitivity to rate/dividend/time assumptions can distort inversion even after an automated relative-spread screen.

The March and June 2027 "curves" each have only three accepted strikes (21,000, 22,500, 24,000). Straight lines between these points should not be read as evidence of a genuinely linear volatility smile.

7. Two illustrative maturity slices

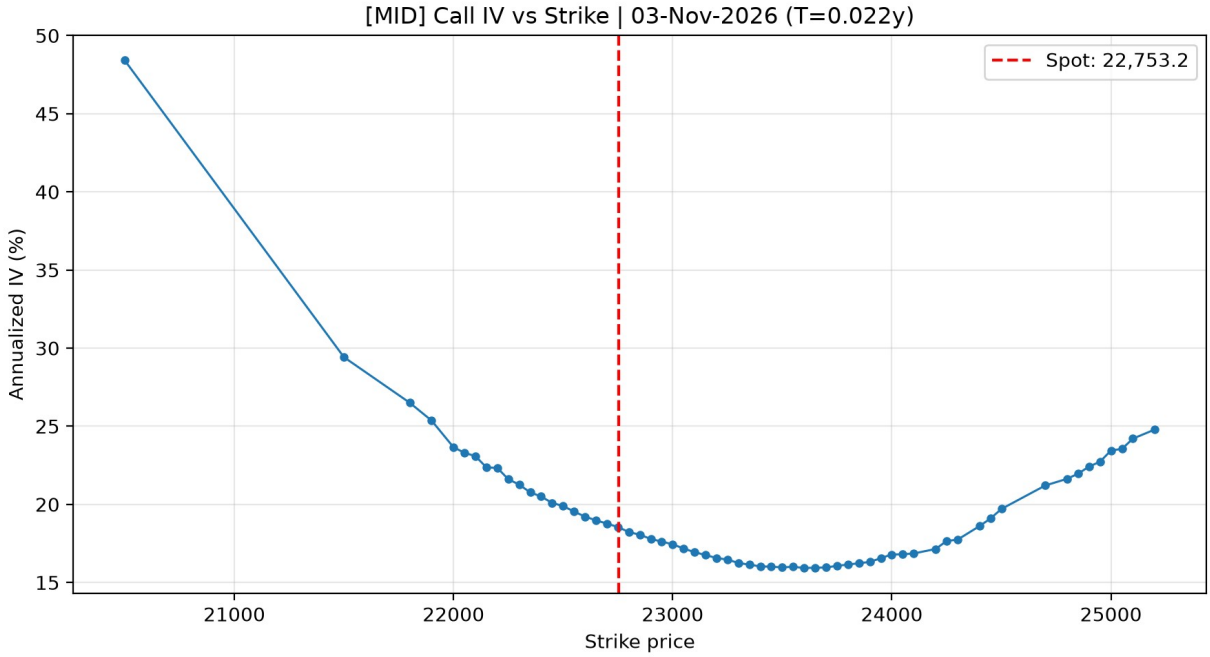


Figure 3. 3 November 2026 midpoint call-IV curve. The leftmost point near strike 20,500 is approximately 48.41% and deserves manual quote review.

The nearest maturity shows a pronounced drop in IV from low strikes toward the money, then an upswing toward the high strikes. Its most extreme left-tail observation passed the present numeric screen, which illustrates why chart acceptance is not proof of clean prices.

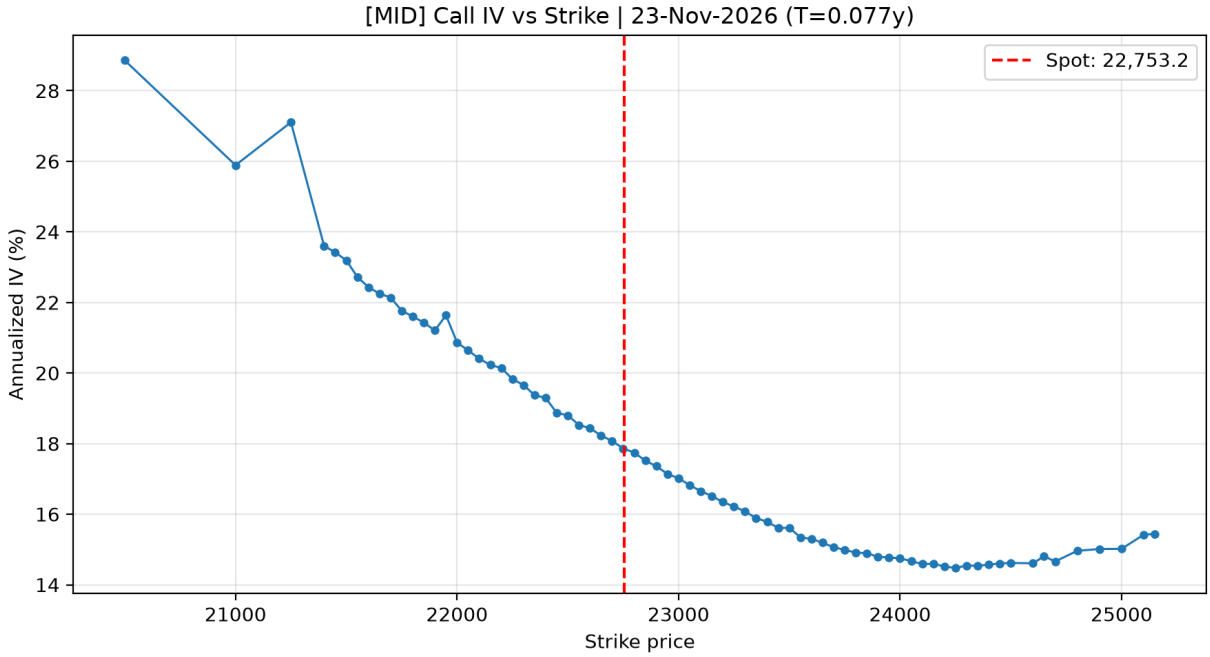


Figure 4. 23 November 2026 midpoint call-IV curve. The broad decline and modest right-tail rise are visible with less extreme low-strike amplification.

8. ATM-strike term structure

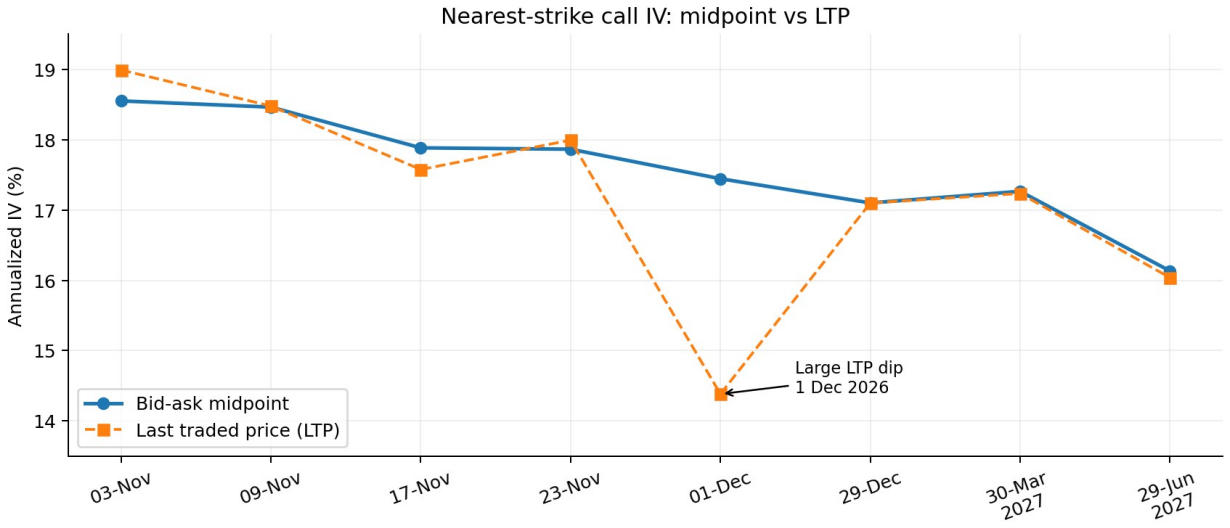


Figure 5. Nearest-available-strike call IV by expiry. The prominent 1 December LTP dip is not present in the midpoint series.

Expiry	ATM strike	Midpoint IV	LTP IV
03-Nov-2026	22,750	18.55%	18.99%
09-Nov-2026	22,750	18.46%	18.48%
17-Nov-2026	22,750	17.88%	17.58%
23-Nov-2026	22,750	17.86%	17.99%
01-Dec-2026	22,750	17.44%	14.38%
29-Dec-2026	22,750	17.10%	17.10%
30-Mar-2027	22,500	17.27%	17.23%
29-Jun-2027	22,500	16.13%	16.04%

The midpoint series generally decreases from 18.55% for 3 November 2026 to 16.13% for 29 June 2027. This is a description of the calculated snapshot, not proof of a stable market expectation or a trading opportunity.

The March and June values use strike 22,500, versus strike 22,750 for earlier expiries. These are nearest-available-strike observations, not a perfectly matched forward-ATM volatility term structure.

9. Midpoint versus last traded price

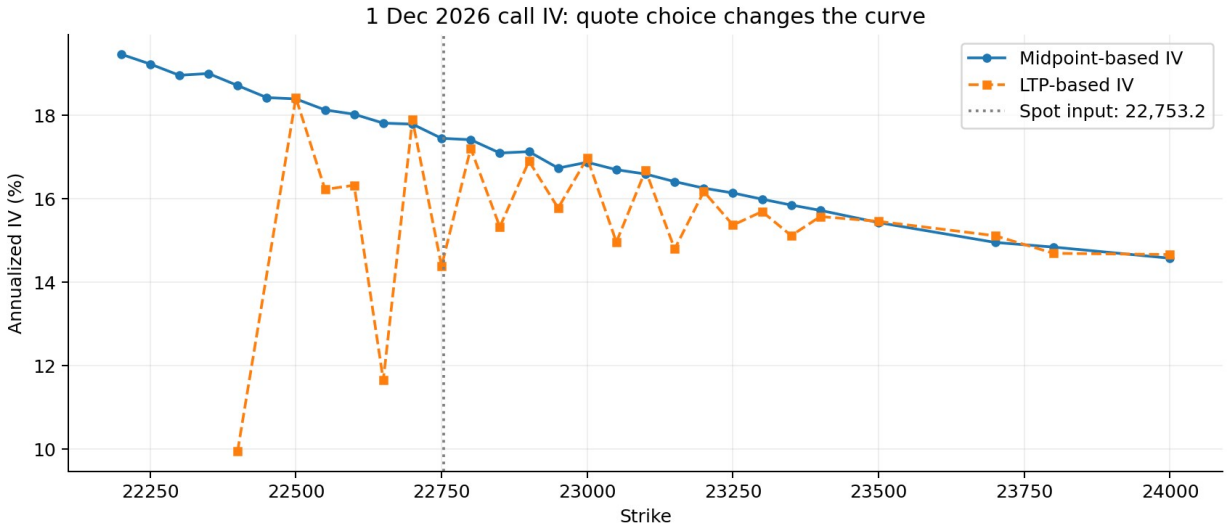


Figure 6. 1 December 2026: several low-strike and ATM LTP IV estimates sit below the midpoint IV curve.

A key market-data discrepancy

1 Dec 2026, strike 22,750	Value
Bid / Ask	INR 536.30 / INR 563.55
Midpoint / LTP	INR 549.925 / INR 463.35
Midpoint IV / LTP IV	17.4449% / 14.3820%
Difference	3.0629 percentage points

The LTP is below the displayed bid, which means the last transaction is not in the contemporaneous bid–ask interval. This is not proof of an erroneous trade: the order book can move after a transaction. It is, however, a strong warning against interpreting the resulting dip as the current volatility surface without further time-aligned data.

Among 312 strike–expiry rows accepted by both approaches, the median absolute IV difference is 0.10 percentage points (mean: 0.57; 90th percentile: 1.25). Yet 95 of 312 accepted LTP observations (30.4%) lie outside the reported bid–ask interval.

10. Three-dimensional volatility visualization

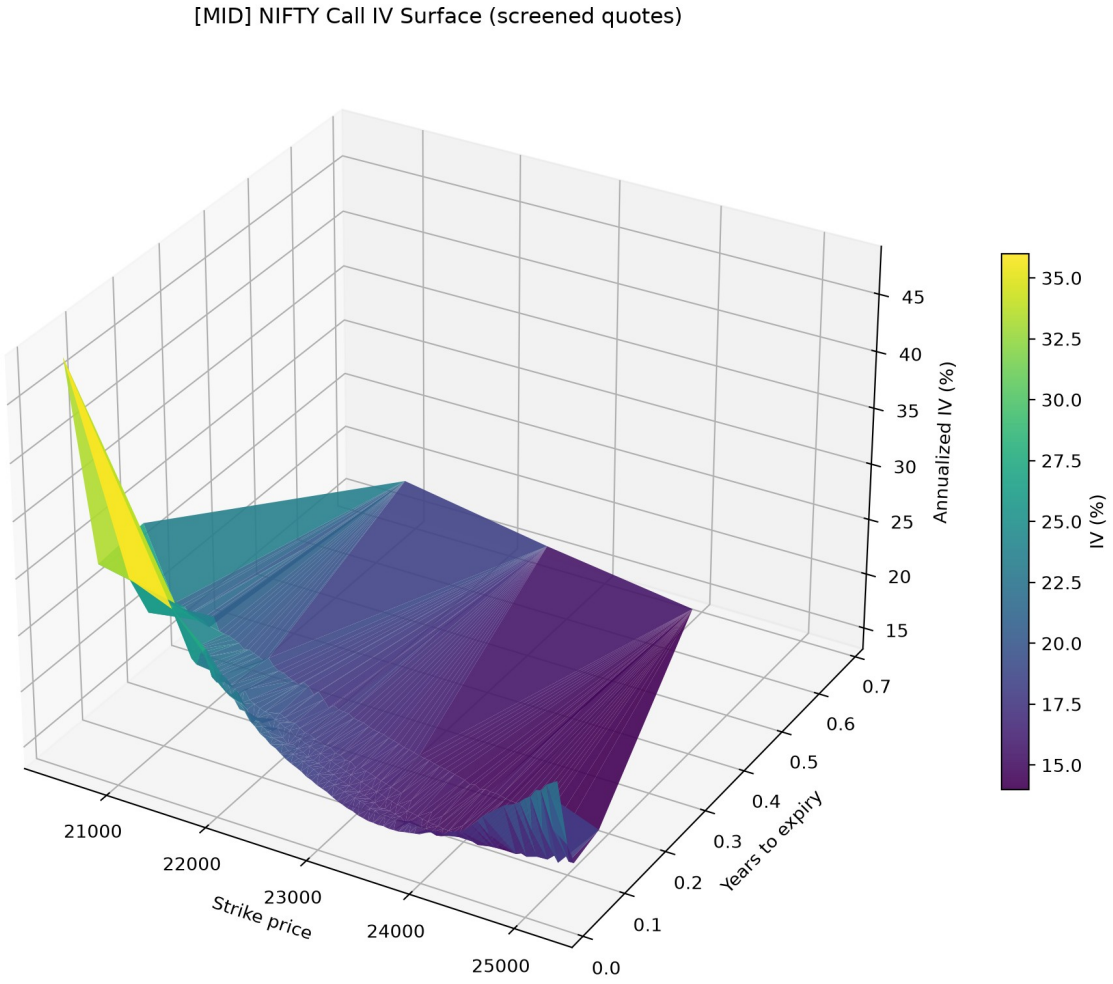


Figure 7. Midpoint-derived strike–maturity–IV triangulation. It is a mesh through screened points, not an arbitrage-free calibrated volatility surface.

A three-dimensional plot puts strike price on one axis, time to expiry on the second, and annualized implied volatility on the vertical axis. In principle, such views help compare skew and maturity dependence simultaneously.

Here the mesh is constructed by triangulation across uneven data coverage. It connects distant observed points, including only three for each long-dated expiry. Broad faces and steep walls partly reflect the interpolation geometry; they should not be read as observed prices for every interior point.

The latest code can also generate a self-contained, rotatable HTML mesh using Plotly. Static PNG output and interactive HTML use the same screened data, but the two renderers may triangulate at slightly different scales.

11. Financial interpretation and practical uses

What the observed skew may mean

The lower-strike side often exhibits higher IV than central/higher strikes. In equity markets this is commonly associated with downside-risk pricing and demand for protective options, but this dataset alone cannot distinguish hedging demand from supply, liquidity or quote-timing effects. Because this project uses calls only, a put-side cross-check would strengthen the interpretation.

Why term structure matters

A cross-expiry IV comparison helps show which maturities have higher model-implied uncertainty. The observed midpoint curve broadly slopes downward, but a single time slice cannot establish a persistent volatility regime. Term-structure values are also sensitive to the choice of spot-ATM versus forward-ATM strike.

Where practitioners use IV and volatility surfaces

- Option quoting and relative-value screening across strikes and expiries.
- Risk management: estimating Vega exposures and examining how option portfolios respond to volatility shocks.
- Portfolio hedging: comparing option costs and sensitivities at different strikes.
- Model calibration: using observed option premiums to fit richer strike- and maturity-dependent models.
- Market commentary: describing the shape of market-implied risk pricing, with appropriate data-quality controls.

An important interpretive boundary

Implied volatility is the volatility parameter implied by a pricing formula at the observed premium. It is not the probability of a market fall, not a guaranteed forecast of realized volatility, and not by itself a “buy” or “sell” signal. A risk-neutral option price does not equal a real-world directional prediction.

12. Limitations and proposed extensions

Limitations that should remain visible

- One download session: no time-series conclusions about how the skew evolves.
- The code fixes spot S = 22,753.2, r = 6.1% and q = 1.52% for every contract; interest rates and dividend yields are proxies, not maturity-specific curves.
- Call-only estimation does not validate implied volatility against liquid out-of-the-money puts.
- The ±12% strike filter and spread/Vega/IV thresholds are practical, chosen heuristics, not exchange or industry-wide standards.
- LTP and bid/ask snapshots are not guaranteed time-synchronized.
- The triangulated surface can display apparent continuity across sparse strike/maturity regions.
- No formal static- or calendar-arbitrage checks; no calibrated smooth arbitrage-free surface.

Logical next research steps

Near term	Optional later
Compare OTM calls and puts	Construct historical/realized volatility from index closes
Add volume/OI-based liquidity diagnostics	Compare IV with trailing or future realized vol, carefully aligned
Use forward moneyness and term-specific rates	Calculate IV–realized spread and robustness intervals
Run put–call parity sanity checks	Build a calibrated or arbitrage-checked smile model

A historical-volatility extension would estimate realized movements from past daily NIFTY returns (for example, sample standard deviation times sqrt(252)). A proper comparison must match the measurement horizon and recognize that implied and realized volatility measure different things. This extension has not been built in this project.

13. Reproducibility and code structure

The repository intentionally processes offline NSE exports; it neither scrapes the NSE website nor needs live connectivity to reproduce the calculations after the CSV files are placed in data/.

Project file	Role
black_scholes.py	Dividend-adjusted call/put pricing, Vega, model price bounds
implied_volatility.py	Newton–Raphson IV inversion with Brent fallback
data/fetch_mid_data.py	Read call bid/ask, LTP and volume; calculate midpoint
data/fetch_ltp_data.py	Read call last-traded prices and quote fields
analyze_options_mid.py	Compute/screen midpoint IV; diagnostics and figures
analyze_options_ltp.py	Compute/screen LTP IV; diagnostics and figures
requirements.txt	NumPy, Pandas, SciPy, Matplotlib, Plotly dependencies

Reproduce locally

1. Create and activate a Python 3.13 virtual environment on Windows: py -3.13 -m venv .venv; . \.venv\Scripts\Activate.ps1.

2. Install dependencies: python -m pip install -r requirements.txt.

3. Put all eight NSE files in data/ under their original expiry-bearing filenames.

4. Verify SPOT_PRICE, SNAPSHOT_DATETIME, RISK_FREE_RATE and DIVIDEND_YIELD in both analysis scripts.

5. Run: python analyze_options_mid.py and python analyze_options_ltp.py.

6. Inspect plots_mid/ and plots_ltp/, including calculated_iv_diagnostics.csv and atm_term_structure.csv. HTML interactive surfaces can be opened in a browser if generated.

For publication, keep raw NSE files and local virtual environments out of Git unless the applicable data-use terms explicitly permit redistribution. Selected static plots can be stored in docs/figures/ and cited by README.md.

14. Conclusion

The project demonstrates an end-to-end empirical IV workflow: offline market-data parsing, dividend-adjusted option pricing, stable numerical inversion, transparent quote-quality screening, cross-strike and cross-expiry analysis, and 3D visualization. Reproduction of the midpoint diagnostic CSV from the original eight raw files confirmed identical plotted/rejected statuses and numerical IV values to machine precision.

The clearest empirical finding is that quote choice changes apparent volatility features. Most accepted midpoint/LTP pairs agree closely, but some LTPs outside the displayed current spread produce severe artificial-looking troughs in the LTP plots. The midpoint-derived nearest-available-strike series is therefore the preferred descriptive view, subject to its own quote-quality and modelling limitations.

Results should be presented as an exploratory analysis of an option-chain snapshot, not as a trading model, forward-volatility forecast, or arbitrage-free fitted volatility surface.

Sources and technical references

- NSE India, NIFTY option-chain downloadable CSV files (eight expiry-specific exports), collected by the project owner on 26 October 2026, 15:40 IST; raw filenames retained in project data folder.
- Project code: black_scholes.py; implied_volatility.py; fetch_mid_data.py; fetch_ltp_data.py; analyze_options_mid.py; analyze_options_ltp.py.
- Project output: calculated_iv_diagnostics.csv and atm_term_structure.csv for both midpoint and LTP versions, as uploaded.
- Black, F. & Scholes, M. (1973), “The Pricing of Options and Corporate Liabilities,” Journal of Political Economy.
- Merton, R. C. (1973), “Theory of Rational Option Pricing,” Bell Journal of Economics and Management Science.
- Hull, J. C., Options, Futures, and Other Derivatives, 11th edition (Pearson).
- SciPy documentation: optimize.brentq; NumPy, Pandas and Matplotlib documentation.

Data-handling note

Raw files provide the observed call bid/ask/LTP and strikes, while diagnostic CSVs provide model-specific IV and screening outputs. All metrics in this report are derived from those files; interpretations of market behaviour are explicitly identified as qualitative and conditional.